24CMIT13 RELATIONAL DATA BASE MANAGEMENT SYSTEMS 4 0 0 4

COURSE OBJECTIVE

- To impart the knowledge of basic concepts of Database systems
- Establish a basic understanding of background materials needed for technical support using MySQL.
- To understand the fundamental concepts of MySql.

UNIT I INTRODUCTION TO DATABASE SYSTEM

Introduction (Basic Concepts: Data, Database, Database systems, Database Management Systems), Characteristics of Database Approach, Advantages of using the DBMS approach Database System Concepts and Architecture: Data Models, Schemas, Instances, the three schema architectures and data independence, Database Languages and interfaces, Database System environment, Centralized and client / Server Architecture for DBMS, Classifications of Database Management Systems.

UNIT II ENTITY RELATIONSHIP MODELLING 12

Entity, Entity Sets, Attributes and keys – Super Key, Primary Key, Candidate Key, Alternate Key, Relationship Types, Composite entities, Weak entity, Subclass, Super class with Attribute inheritance, Generalization and Specialization. Relational Database design by ER and EER to Relational Mapping, Mapping EER model construct to Relations.

UNIT III DATABASE DESIGN 12

Informal Design Guidelines for Relational Schema, Functional Dependencies, Normal Forms based on Primary keys, General definitions of 1NF, 2NF and 3NF, Boyce-Codd Normal Forms (BCNF), Multi-valued Dependency and Fourth Normal Form.

UNIT IV SQL & MySQL 12

SQL: Commands – Data types – DDL - Selection, Projection, Join and Set Operations – Aggregate Functions – DML – Modification - Truncation - Constraints – Subquery.

Basics of MySQL

Features of MySQL, Data types, Difference between SQL and MySQL, Integrity

Constraints-(Primary key, Foreign key, unique key, not Null, default, check)DDL, DML, DCL, TCL Commands, Select Statement with Clauses-Where, Having, ORDERBY and GROUPBY.

UNIT V FUNCTIONS AND JOINS IN MYSQL 12

Functions in MySQL-Aggregate functions, String Functions, Math Functions, Date and Time Functions, Joins and Sub queries in MySQL, MySQL control statements and stored procedures, MySQL Triggers.

Total Hours 60

Course Outcome

At the end of this course, students will be able to:

CO-1: Grasp the foundational principles of database systems, including their purpose, components, and importance in data management.

CO-2: Understand the concepts of Entity Relationship Model

CO-3: Able to understand and design simple ER model

CO-4: Apply normalization techniques to ensure database efficiency.

CO-5: Apply normalization techniques, including identifying functional dependencies and normal forms, to design efficient and well-structured databases.

CO-6: Analyze various types of SQL statements and their applications.

CO-7: Understand and describe the basic concepts of Mysql

CO-8: Able to write queries using Mysql

TEXT BOOKS

1. Ramez Elmsari, Shamkant B Navathe, “Fundamentals of Database Systems”, Pearson Education, 7th Edition.

2. VaswaniVikram-MySQL(TM): The Complete Reference - The Complete Reference

Paperback.

REFERENCE BOOKS

1. Abraham Silberchatz, Henry F. Korth, S. Sudarshan, “Database System Concepts”, McGrawHill 2019, 7th Edition.

2. Alexis Leon & Mathews Leon, “Fundamentals of DBMS”, Vijay Nicole Publications 2014, 2nd Edition.

WEB REFERENCES

1. https://docs.oracle.com/en/database/index.html

2. https://docs.oracle.com/database/121/SQLRF/toc.htm

WEB SITES

1. www.w3schools.com

2. www.tutorialspoint.com

3. www.javapoint.com

WEB SOURCES

1. www.guru99.com

2. www.youtube.com/watch?v=D6Q_wHrzxDs

24PMIT12 RELATIONAL DATA BASE MANAGEMENT SYSTEMS LAB 0 0 4 2

COURSE OBJECTIVES:

This course gives practical training in design and implementation of relational data bases for the selected set of problems.

Write a program for

1. Table creation, renaming a table, copying another table, dropping a table.
2. DDL commands
3. DML commands
4. Joins and subqueries.
5. Select statements with all clauses /options
6. Index and view.
7. Application development: students mark sheet processing, pay roll processing.
8. Functions in MySQL.
9. MySQL control statements.

10. MySQL triggers. Total Hours: 30

COURSE OUTCOMES:

At the end of this course, the student will be able to:

CO-1: Creating a table using basic commands.

CO-2: Apply DDL, DML statement

CO-3: Apply various types of joins in tables.

CO-4: Apply various types of joins in tables.

CO-5: Evaluate the functionalities of trigger.

CO-6: Analyse different types of built-in function.

CO-7: Able to write queries using Mysql

WEB REFERENCES:

1. https://docs.oracle.com/en/database/index.html

2. https://docs.oracle.com/database/121/SQLRF/toc.html

WEBSITES:

1. www.w3schools.com

2. www.tutorialspoint.com

3. www.javapoint.com